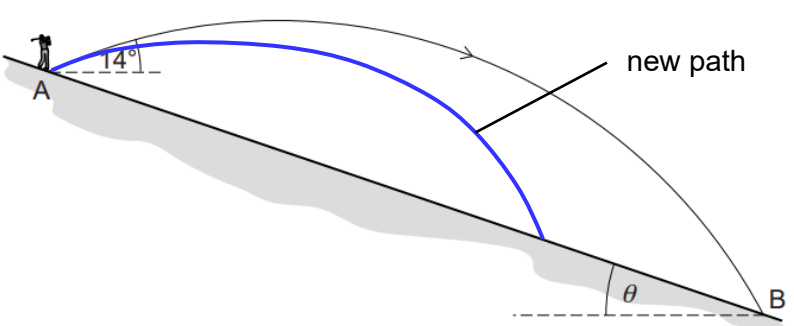


No.	Solution	Remarks
1(a)(i)	horizontal component $= 67 \cos 14^\circ$ $= 65.01$ $\approx 65.0 \text{ m s}^{-1}$ vertical component $= 67 \sin 14^\circ$ $= 16.21$ $\approx 16.2 \text{ m s}^{-1}$	 [1] for correct answer [1] for correct answer
1(a)(ii)	horizontal displacement $= 65.01 \times 4.9$ $= 318.5$ $\approx 319 \text{ m}$	[1] for correct substitution and answer
1(a)(iii)	Using $s_y = u_y t + \frac{1}{2} a_y t^2$, Taking downward as positive, $s_y = (-16.21)(4.9) + \frac{1}{2}(9.81)(4.9)^2$ $= 38.34$ $\approx 38.3 \text{ m}$	 [1] for correct substitution [1] for correct answer
1(a)(iv)	angle of the slope to the horizontal $\theta = \tan^{-1} \left(\frac{38.34}{318.5} \right)$ $= 6.864$ $\approx 6.86^\circ$	 [1] for correct substitution [1] for correct answer
1(b)(i)	 [1] for correct new path Note: new path under present path	

No.	Solution	Remarks
1(b)(ii)1.	The path of the ball at the start is determined by the direction of the force of the club acting on the ball or air resistance has acted only for a short period of time.	[1] for correct answer
1(b)(ii)2.	The horizontal velocity is much reduced by air resistance.	[1] for correct answer
2(a)(i)		